



Universidad Nacional
Autónoma de México

Facultad de Ciencias, 2027-1

Bases de Datos



Practica 01:
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Fecha de entrega: 06 de Septiembre de 2026

1. **Sistema Operativo y Versión:**

Linux 7.1.13-200.fc44.x86 64

2. **Distribución:**

Fedora 44

3. **Versión de la instalación**

- Docker:
- PostgreSQL
- DBeaver

4. **Tiempo Requerido**

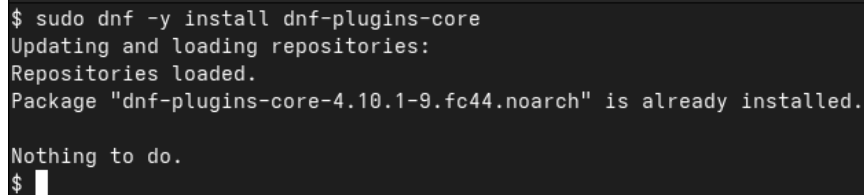
70 minutos

5. **Explicación del paso a paso que realizaste con sus respectivas capturas de pantallas.**

Instalación de Docker

1. Se descargan los paquetes necesarios para agregar el repositorio de Docker (Ya estaban instalados).

```
sudo dnf -y install dnf-plugins-core
```



```
$ sudo dnf -y install dnf-plugins-core
Updating and loading repositories:
Repositories loaded.
Package "dnf-plugins-core-4.10.1-9.fc44.noarch" is already installed.

Nothing to do.
$
```

2. Se agrega el repositorio de Docker para realizar la instalación.

```
sudo dnf config-manager --add-repo https://download.
docker.com/linux/fedora/docker-ce.repo
```

```
$ sudo dnf config-manager addrepo --from-repofile=https://download.docker.com/linux/fedora/docker-ce.repo
https://download.docker.com/linux/fedora 100% | 2.6 KiB/s | 811.0 B | 00m00s
$
```

3. Se instalan paquetes principales de Docker.

```
sudo dnf install docker-ce docker-ce-cli containerd.io
docker-buildx-plugin docker-compose-plugin
```

```
$ sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin do
cker-compose-plugin
Updating and loading repositories:
  Docker CE Stable - x86_64          100% | 36.2 KiB/s | 13.9 KiB | 00m00s
Repositories loaded.
Package                               Arch    Version                Repository    Size
Installing:
  containerd.io                       x86_64  0:2.3.4-2.fc44         docker-ce-st 112.7 MiB
  docker-buildx-plugin                x86_64  0:0.37.0-1.fc44        docker-ce-st 69.7 MiB
  docker-ce                           x86_64  3:29.8.0-1.fc44        docker-ce-st 110.3 MiB
  docker-ce-cli                       x86_64  1:29.8.0-1.fc44        docker-ce-st 36.5 MiB
  docker-compose-plugin               x86_64  0:5.5.1-1.fc44         docker-ce-st 30.9 MiB
Installing dependencies:
  libcgroupr                          x86_64  0:3.0-10.fc44          fedora       161.6 KiB
Installing weak dependencies:
  docker-ce-rootless-extras          x86_64  0:29.8.0-1.fc44        docker-ce-st 20.1 MiB

Transaction Summary:
  Installing:          7 packages

Total size of inbound packages is 99 MiB. Need to download 99 MiB.
After this operation, 380 MiB extra will be used (install 380 MiB, remove 0 B).
Is this ok [y/N]: y
```

4. Se activa la maquina de Docker.

```
sudo systemctl enable --now docker
```

```
complete!
$ sudo systemctl enable --now docker
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' ->
'/usr/lib/systemd/system/docker.service'.
+
```

5. Se revisa el estado actual de Docker.

```
sudo systemctl status docker
```

```

$ sudo systemctl status docker
[sudo] password for diegocist:
• docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: d>
   Drop-In: /usr/lib/systemd/system/service.d
           └─10-timeout-abort.conf
   Active: active (running) since Sun 2026-09-06 18:03:35 CST; 1h 22min ago
   Invocation: 5982ee7af86148f8a0169dbf8c420ea9
   TriggeredBy: • docker.socket
     Docs: https://docs.docker.com
    Main PID: 21592 (dockerd)
      Tasks: 13
     Memory: 46.5M (peak: 119.5M, swap: 7.8M, swap peak: 8.2M)
        CPU: 1.378s
     CGroup: /system.slice/docker.service
            └─21592 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/con>

Sep 06 18:03:34 fedora dockerd[21592]: time="2026-09-06T18:03:34.144812757-06:0>
Sep 06 18:03:34 fedora dockerd[21592]: time="2026-09-06T18:03:34.291556491-06:0>
Sep 06 18:03:34 fedora dockerd[21592]: time="2026-09-06T18:03:34.291782355-06:0>
Sep 06 18:03:35 fedora dockerd[21592]: time="2026-09-06T18:03:35.505317787-06:0>
Sep 06 18:03:35 fedora dockerd[21592]: time="2026-09-06T18:03:35.514308031-06:0>
Sep 06 18:03:35 fedora dockerd[21592]: time="2026-09-06T18:03:35.514344086-06:0>
Sep 06 18:03:35 fedora systemd[1]: Started docker.service - Docker Application >
Sep 06 18:04:42 fedora dockerd[21592]: time="2026-09-06T18:04:42.554215685-06:0>
Sep 06 18:04:49 fedora dockerd[21592]: time="2026-09-06T18:04:49.124098073-06:0>
Sep 06 18:04:49 fedora dockerd[21592]: time="2026-09-06T18:04:49.583159437-06:0>
$

```

6. Se verifica que la instalación fue realizada correctamente

```
sudo docker run hello-world
```

```

$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
d5e71e642bf5: Download complete
4f55086f7dd0: Pull complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

```

Contenedor para el SMBD PostgreSQL

1. Se descarga la imagen de Postgres para crear el contenedor.

```
sudo docker pull postgres
```

```
$ sudo docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
19a25d2b9e16   postgres  "docker-entrypoint.s..." 18 seconds ago Up 9 seconds
fb763c450b6b   hello-world "/hello"                21 minutes ago Exited (0)
ade233450783   hello-world "/hello"                2 hours ago    Exited (0)
```

2. Se crea el contenedor.

```
sudo docker run -d \
--name postgres \
-e POSTGRES_PASSWORD=mysecretpassword \
-p 127.0.0.1:5432:5432 \
-v postgres_data:/var/lib/postgresql \
postgres
```

3. Se revisa el container ID del contenedor creado.

```
sudo docker ps -a
```

```
$ sudo docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
19a25d2b9e16   postgres  "docker-entrypoint.s..." 18 seconds ago Up 9 seconds
fb763c450b6b   hello-world "/hello"                21 minutes ago Exited (0)
ade233450783   hello-world "/hello"                2 hours ago    Exited (0)
```

4. Se prueban los comandos para iniciar y detener el contenedor.

```
sudo docker start 19a25d2b9e16
sudo docker stop 19a25d2b9e16
```

```
$ sudo docker start 19a25d2b9e16
19a25d2b9e16
$ sudo docker stop 19a25d2b9e16
19a25d2b9e16
$
```

5. Se contacta a PostgreSQL para usar psql.

```
docker run -it --link postgres:postgres postgres psql -h
postgres -U postgres
```

```
$ sudo docker run -it --link postgres:postgres postgres psql -h postgres -U pos
tgres
WARNING: Links on the default bridge network are deprecated and will be removed
in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

postgres=# \q
```

Instalacion Database Tools

PgAdmin4

1. Se agrega el repositorio de PgAdmin4 para realizar la instalación.

```
docker run -it --link postgres:postgres postgres psql -h
postgres -U postgres
```

```
$ sudo rpm -i https://ftp.postgresql.org/pub/pgadmin/pgadmin4/yum/pgadmin4-fedora-repo-2-1.noarch.rpm
[sudo] password for diegocist:
warning: /var/tmp/rpm-tmp.VcEl3m: Header OpenPGP V4 RSA/SHA256 signature, key ID 8881b2a8210976f2: NOKEY
```

2. Se realiza la instalacion de PgAdmin4

```
sudo dnf install pgadmin4-desktop
```

```
$ sudo dnf install pgadmin4-desktop
Updating and loading repositories:
pgadmin4 100% | 15.5 KiB/s | 3.8 KiB | 00m00s
Importing OpenPGP key 0x210976F2:
UserID : "Package Manager (Package Signing Key) <packages@pgadmin.org>"
Fingerprint: E8697E2EEF76C02D3A6332778881B2A8210976F2
From : file:///etc/pki/rpm-gpg/PGADMIN_PKG_KEY
Is this ok [y/N]: y
The key was successfully imported.
pgadmin4 100% | 192.2 KiB/s | 39.2 KiB | 00m00s
Repositories loaded.
Package Arch Version Repository Size
Installing:
pgadmin4-desktop x86_64 0:9.17-1.fc44 pgAdmin4 314.7 MiB
Installing dependencies:
libpq5 x86_64 0:18.4-1PGDG.f44 pgAdmin4 1.1 MiB
replacing libpq x86_64 0:18.0-4.fc44 fedora 1.0 MiB
pgadmin4-server x86_64 0:9.17-1.fc44 pgAdmin4 371.0 MiB

Transaction Summary:
Installing: 3 packages
Replacing: 1 package

Total size of inbound packages is 184 MiB. Need to download 184 MiB.
After this operation, 686 MiB extra will be used (install 687 MiB, remove 1 MiB).
Is this ok [y/N]: y
[1/3] libpq5-0:18.4-1PGDG.f44.x86_64 100% | 208.9 KiB/s | 291.0 KiB | 00m01s
[2/3] pgadmin4-desktop-0:9.17-1.fc44.x86_64 29% [=====] | 5.3 MiB/s | 26.6 MiB | 00m12s
[2/3] pgadmin4-desktop-0:9.17-1.fc44.x86_64 100% | 2.8 MiB/s | 91.7 MiB | 00m33s
[3/3] pgadmin4-server-0:9.17-1.fc44.x86_64 100% | 1.7 MiB/s | 91.8 MiB | 00m54s
```

